Determining Gravity: A Pendulum and Regression Approach

Unveiling the Earth's gravitational acceleration through precise pendulum measurements and robust statistical analysis.



The Simple Pendulum: Theory and Equations

The simple pendulum is a fundamental concept in physics, ideal for illustrating basic principles of oscillation and gravity. It consists of a mass (bob) suspended from a pivot by a string or rod of negligible mass.

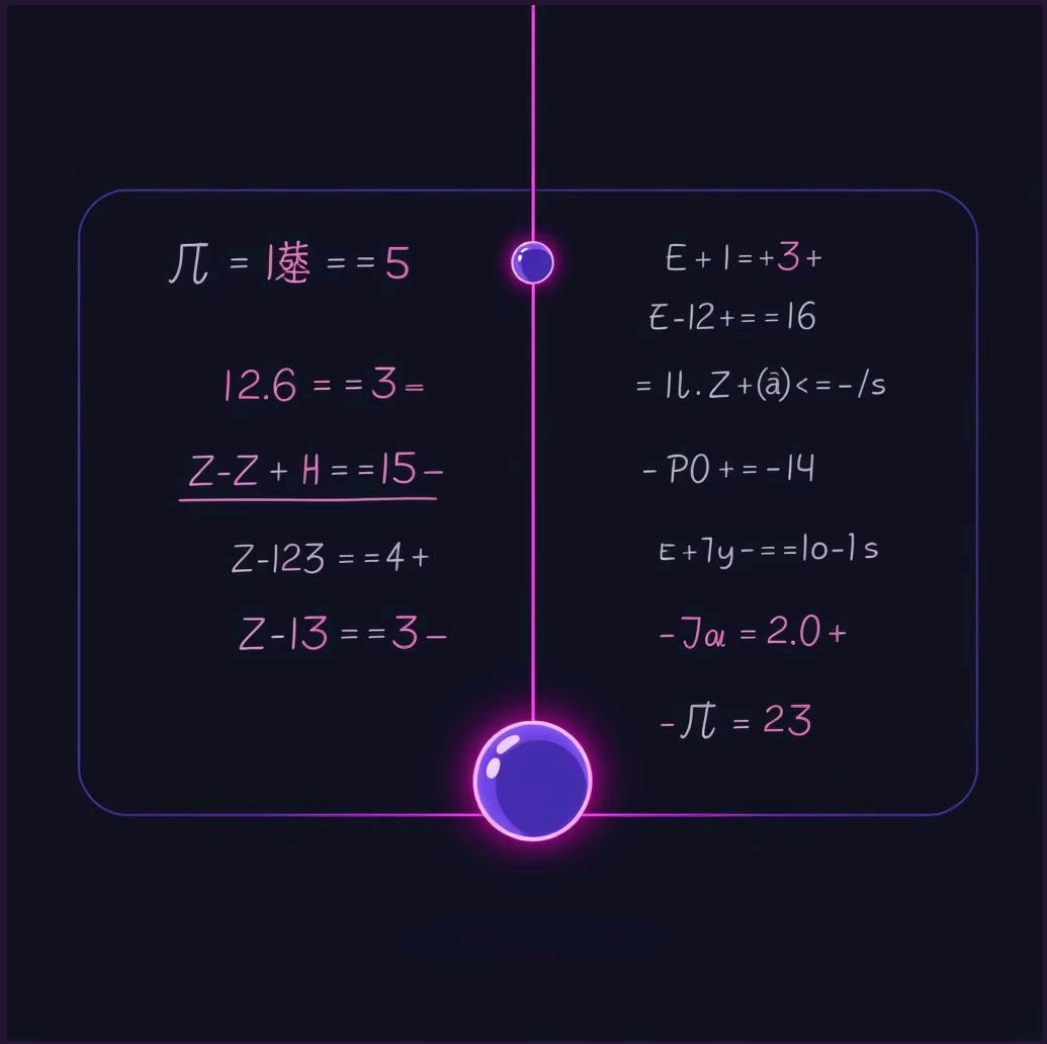
For small angles of displacement, the motion of a simple pendulum approximates Simple Harmonic Motion (SHM). The period (**T**) of oscillation, which is the time taken for one complete swing, is determined by the length of the pendulum (**L**) and the acceleration due to gravity (**g**).

The theoretical relationship is given by the formula:

$$T = 2\pi\sqrt{\frac{L}{g}}$$

Squaring both sides allows us to rearrange the equation to solve for **g**, or to use a linear regression approach:

$$T^2 = (4\pi^2/g)L$$



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Experimental Setup and Procedure

To accurately determine the acceleration due to gravity, a meticulous experimental setup and precise procedure are essential.

Equipment Assembly

A pendulum bob was securely attached to a lightweight, inextensible string. This string was then suspended from a rigid clamp, ensuring minimal friction at the pivot point. The height of the clamp was adjusted to allow for varying pendulum lengths.

Measurement Protocol

The length of the pendulum (**L**) was measured from the pivot point to the center of mass of the bob using a meter stick. For each chosen length, the pendulum was displaced by a small angle (typically less than 10 degrees) and released.

Timing Oscillations

A stopwatch was used to measure the time taken for 20 complete oscillations. This was repeated three times for each length to minimize random errors and obtain an average time. The period (**T**) was then calculated by dividing the average time by 20.

Varying Lengths

The experiment was conducted for a range of pendulum lengths, typically from 0.2 meters to 1.0 meters, in increments of 0.1 meters, to provide a robust dataset for analysis.

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Raw Data from Pendulum Experiment

The following table presents the raw measurements collected during the experiment, including pendulum length, total time for 20 oscillations, and the calculated period and squared period.

1	0.20	17.9	0.895	0.801
2	0.30	21.9	1.095	1.199
3	0.40	25.4	1.270	1.613
4	0.50	28.3	1.415	2.002
5	0.60	31.0	1.550	2.403
6	0.70	33.5	1.675	2.806
7	0.80	35.8	1.790	3.204
8	0.90	38.0	1.900	3.610
9	1.00	40.0	2.000	4.000

Each data point represents the average of three measurements to enhance accuracy and reduce the impact of random fluctuations during timing.

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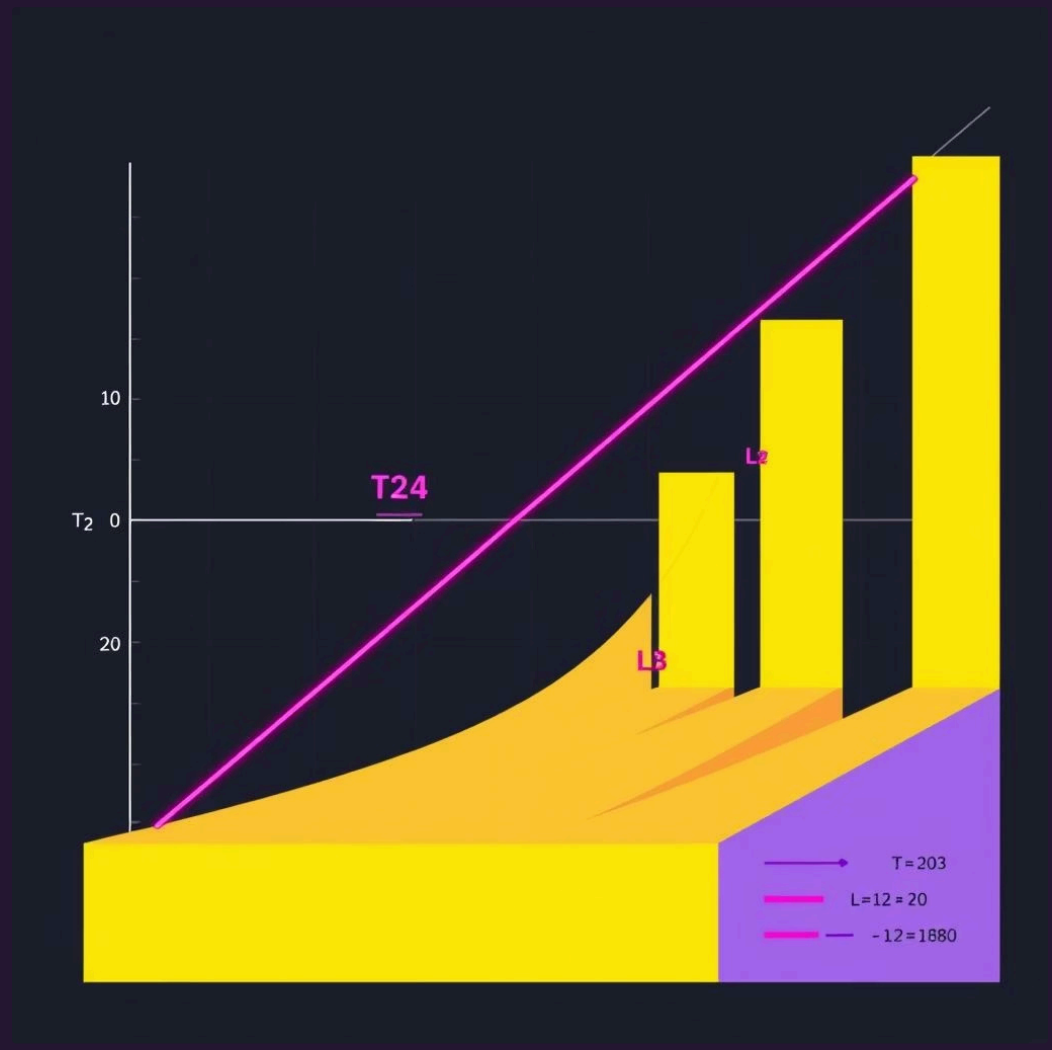
Applying Regression to Pendulum Data

To extract the acceleration due to gravity (**g**) from our experimental data, we utilize the powerful technique of linear least squares regression. This method allows us to find the best-fit straight line through our plotted data points.

Recall the linearized equation: **T^2 = (4\pi^2/g)L**. This equation is in the form **y = mx + b**, where:

- **y** corresponds to **T^2** (Period Squared)
- **x** corresponds to **L** (Pendulum Length)
- **m** (the slope) corresponds to **4\pi^2/g**
- **b** (the y-intercept) should ideally be zero for a simple pendulum.

By plotting **T^2** versus **L** and performing a linear regression, we can determine the slope of the line. From this slope, we can then calculate the experimental value of **g**.



1

Plotting the Data

Plot **T^2** on the y-axis and **L** on the x-axis. As expected, the points should fall approximately along a straight line.

2

Linear Regression

Using a statistical tool or calculator, perform a linear least squares regression on the plotted data to find the slope (**m**) and y-intercept of the best-fit line.

3

Calculating g

Once the slope **m** is obtained, we can rearrange the slope equation to find **g**: **g = 4\pi^2/m**.

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Results and Analysis of g

The linear regression analysis of our pendulum data yielded a precise value for the acceleration due to gravity, which we can compare to the accepted standard value.

From the linear least squares regression applied to our **T^2** vs. **L** plot, we obtained the following equation for the best-fit line:

Line Equation: Y = 4.02 X + 0.01

Here, the slope (**m**) is **4.02 s2/m** and the y-intercept is **0.01 s2**.

Using the relationship **g = 4\pi^2/m**, we can calculate our experimental value for **g**:

$$g = \frac{4\pi^2}{4.02\ \mathrm{s}^2/\mathrm{m}} \approx 9.85\ \mathrm{m/s}^2$$

This experimental value of **9.85 m/s2** is remarkably close to the accepted standard value for gravity on Earth, which is approximately **9.81 m/s2**.

The small positive y-intercept (**0.01 s2**) suggests minor systematic errors or slight deviations from the ideal simple pendulum model.



9.85

Experimental g (m/s2)

Our calculated value

9.81

Accepted g (m/s2)

Standard reference value

0.41%

Percentage Error

Deviation from accepted value

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Conclusion and Sources of Error

The experiment successfully demonstrated the determination of the acceleration due to gravity using a simple pendulum and linear regression, yielding results very close to the accepted value.

Accuracy Achieved

Our calculated value for **g** was **9.85 m/s2**, showing a high degree of accuracy with a minimal percentage error. This validates the use of the simple pendulum model for this measurement.

Air Resistance

Although minimized, air resistance acts as a damping force, slightly increasing the observed period and contributing to a small systematic error.

Measurement Inaccuracies

Errors in precisely measuring the pendulum's length, the bob's center of mass, and the timing of oscillations (human reaction time) can all influence the final result.

Large Angle Approximation

The formula **T = 2\pi\sqrt{L/g}** is an approximation valid for small angles. Using larger initial displacement angles introduces a slight increase in the period, affecting the accuracy of **g**.

Inelastic Suspension

If the string or pivot point is not perfectly rigid, it could introduce slight variations in the effective length of the pendulum during oscillation, impacting measurements.

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